

Checkpoint 6 — Score Fusion & Decision Service

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1 Score Fusion Design

1.1 Problem Statement

The Autoencoder and LSTM produce anomaly scores on fundamentally different scales. The AE outputs mean squared reconstruction error (test p99 = 0.643), while the LSTM outputs next-event prediction error (test p99 = 3.746). A weighted combination of raw scores is meaningless without normalization. Additionally, the two models have different strengths depending on client maturity: the AE scores individual events in isolation (effective from the first transaction), while the LSTM requires sufficient history to form meaningful sequences.

1.2 Normalization: Percentile Ranking

Each model’s raw score is normalized via percentile rank against a saved reference distribution of normal training scores.

Procedure:

1. After training, run inference on the normal training set (AE: 142,254 events; LSTM: 99,454 normal training sequences).
2. Sort the resulting scores and save as `models/ae_ref_scores.npy` and `models/lstm_ref_scores.npy`.
3. At scoring time, given a raw score s , compute its percentile via binary search (`np.searchsorted`) against the sorted reference array:

$$p = \frac{\text{searchsorted}(s, \text{ref_sorted})}{|\text{ref_sorted}|}$$

Both normalized scores now lie in $[0, 1]$ with identical semantics: “what fraction of normal behavior scored below this event.” A score of 0.99 means the event is more abnormal than 99% of normal training events.

Reference distributions are recomputed alongside every model retrain (weekly batch or drift-triggered), ensuring alignment between model weights and score calibration.

Table 1: Reference distribution statistics (normal training scores)

Model	p50	p95	p99
Autoencoder (142,254 scores)	0.1150	0.3782	0.6429
LSTM (99,454 scores)	0.3701	1.3184	3.7459

1.3 Maturity-Weighted Fusion

The two percentile scores are combined via a sigmoid-weighted average anchored on client maturity:

$$w_{\text{lstm}} = \sigma\left(\frac{d - \mu}{k}\right) = \frac{1}{1 + e^{-(d - \mu)/k}}$$

where $d = \text{days_since_account_opening}$, $\mu = 90$ (midpoint), $k = 15$ (steepness).

$$\text{score}_{\text{fused}} = (1 - w_{\text{lstm}}) \cdot p_{\text{ae}} + w_{\text{lstm}} \cdot p_{\text{lstm}}$$

Behavior:

- $d = 30 \text{ days} \Rightarrow w_{\text{lstm}} \approx 0.02$ (AE-dominated)

- $d = 90$ days $\Rightarrow w_{\text{lstm}} = 0.50$ (equal weight)
- $d = 150$ days $\Rightarrow w_{\text{lstm}} \approx 0.98$ (LSTM-dominated)

When no LSTM sequence is available (client has fewer than `seq_len=10` prior events), the system falls back to $w_{\text{lstm}} = 0$ (pure AE).

1.4 Account Opening Date

The `account_opening_date` field was added to `clients.csv` via a post-hoc script (`scripts/add_account_opening_date.py`) with archetype-specific ranges to ensure realistic maturity variation:

Table 2: Account opening date ranges by archetype

Archetype	Range before sim start	Rationale
Retiree	5–20 years	Rare bank switching
Salaried	2 weeks–10 years	Mix including genuinely new
Student	1 month–3 years	Young, recent accounts
Small business	2 weeks–10 years	Mix including genuinely new
Big business	2–15 years	Established entities

Result: 190 clients under 90 days (AE-dominated), 432 under 180 days, retirees all 5+ years (LSTM-dominated).

1.5 Fusion Results

Table 3: Model comparison on test set (36,059 events, 175 anomalies)

Model	AUC	Coverage
AE only	0.9935	36,059 / 36,059 (100%)
LSTM only	0.9430	35,415 / 36,059 (98.2%)
Fused	0.9725	36,059 / 36,059 (100%)

The fused AUC (0.9725) is lower than the AE alone on this synthetic dataset because anomalies produce strong per-event signals that favor the point model. In production, subtle sequential patterns (gradual behavioral drift, multi-day smurfing campaigns) will not produce extreme single-event reconstruction errors, and the LSTM’s sequential awareness will earn its weight. The fusion design is validated as mechanically correct with smooth cold-start degradation.

Sigmoid weight distribution on test set: 234 events AE-dominated ($w < 0.1$), 469 mixed ($0.1 \leq w \leq 0.9$), 35,356 LSTM-dominated ($w > 0.9$).

2 Decision Service

2.1 Architecture

The Decision Service is a stateless engine consuming scored events and producing alert tier assignments. It receives a **DecisionInput** (fused score + raw event fields + enriched features + client profile context + risk history) and returns a **Decision** (tier + reasons + regulatory flags).

Three layers are applied sequentially:

1. **Layer 1 — Score-based tier assignment:** calibrated thresholds from the validation set
2. **Layer 2 — Regulatory rule escalation:** deterministic LAB/FT rules that can only *escalate* the tier, never de-escalate
3. **Layer 3 — Risk history adjustment:** recidivist escalation and false-positive dampening based on supervisor feedback

2.2 Threshold Calibration

Thresholds were calibrated on the validation set (36,548 events, 312 anomalies) by analyzing the fused score distribution of normal vs. anomalous events and targeting operationally sustainable daily alert volumes.

Table 4: Calibrated score thresholds

Threshold	Tier boundary	Value	Design rationale
T_1	INFO $\rightarrow$ REVIEW	0.95	p95 of normal scores; supervisor checks when free
T_2	REVIEW $\rightarrow$ ALERT	0.99	High confidence; investigate now
T_3	ALERT $\rightarrow$ BLOCK	0.999	Near-certainty; stop the transaction

2.3 Regulatory Rules (Layer 2)

Eight context-aware rules. All rules append to a **regulatory_flags** list for audit traceability. The **max(tier, target)** pattern guarantees escalation only.

#	Rule	Condition	Escalates to	Context guard
1	Reporting threshold (BCT)	amount $\geq$ 10,000 DT	ALERT	Only if z_amount $\geq$ 2
2	Smurfing band	amount $\in$ [8k, 10k) DT & near_threshold_count_7d $\geq$ 2	ALERT	—
3	Cheque ceiling (Law 41-2024)	cheque amount $\in$ [25k, 30k) DT	REVIEW	—
4	Passeur de fonds (CTAF)	occasional client & versement & tx_count_24h $\geq$ 5	ALERT	regulatory_client_type = occasional
5	Large virement + new beneficiary	virement $\geq$ 5,000 DT & new beneficiary	REVIEW	—

#	Rule	Condition	Escalates to	Context guard
6	Duplicate transaction	has_duplicate_recent = true	REVIEW	—
7	Round amount structuring	round amount $\geq$ 5,000 DT & cumulative_24h $\geq$ 15,000 DT	REVIEW	—
8	Frequency burst	tx_count_24h $\geq$ expected_daily $\times$ 5	REVIEW	Per-client expected rate from profile

Context-aware design: Rule 1 always logs the regulatory flag (audit trail requirement) but only escalates the tier when the amount is statistically unusual for the specific client ($z_amount \geq 2$). This prevents big business clients performing routine high-value transactions from flooding the ALERT queue. Rule 8 uses the client’s personal expected daily transaction rate (derived from their sampled frequency parameters) rather than a fixed count, so the same rule behaves differently for retirees (~ 0.1 ops/day) vs. big business (~ 2.1 ops/day).

2.4 Risk History Adjustment (Layer 3)

- **Recidivist escalation:** ≥ 3 confirmed alerts in 30 days $\Rightarrow$ tier $+= 1$ (capped at BLOCK)
- **False-positive dampening:** ≥ 3 rejected alerts in 30 days & tier = REVIEW & no regulatory flags $\Rightarrow$ REVIEW $\rightarrow$ INFO
- **Cold-start conservatism:** maturity_status = cold & fused score ≥ 0.90 & tier = INFO $\Rightarrow$ escalate to REVIEW

Dampening has a safety guard: it only touches REVIEW (never ALERT or BLOCK), and it does not fire if any regulatory rule triggered. Risk history fields are all zeros on first run (no prior alert history exists); the adjustment layer becomes active once the supervisor feedback loop is operational.

2.5 End-to-End Results

Table 6: Decision Service output on test set (36,059 events, 175 anomalies)

Tier	Events	Anomalies caught	Precision
INFO	28,092 (77.9%)	11	0.0%
REVIEW	7,082 (19.6%)	26	0.4%
ALERT	730 (2.0%)	28	3.8%
BLOCK	155 (0.4%)	110	71.0%

Missed anomalies (11): 8 frequency_burst, 2 cumulative_threshold, 1 timing_anomaly. Frequency burst remains the hardest scenario because the anomaly manifests as event *count* rather than any per-event feature deviation. The frequency burst rule (Rule 8) recovered 4 additional frequency_burst events that the ML models missed, improving detection from 91.4% to 93.7%.

Table 7: Operational metrics

Metric	Value
Overall detection rate (REVIEW+ALERT+BLOCK)	93.7%
BLOCK precision	71.0%
Daily REVIEW volume (bank-wide / per branch)	262 / ~ 1.7
Daily ALERT volume (bank-wide / per branch)	27 / ~ 0.18
Daily BLOCK volume (bank-wide)	5.7
Events with regulatory flags	10,349

3 Artifacts Produced

File	Description
<code>src/scoring/fusion.py</code>	ScoreFusion class: loads models + reference distributions, percentile normalization, sigmoid maturity weight, single-event and batch scoring
<code>src/decision/decision.py</code>	DecisionService class: 3-layer stateless engine (score thresholds + 8 regulatory rules + risk history adjustment). Includes DecisionInput/Decision dataclasses
<code>models/ae_ref_scores.npy</code>	Sorted AE reference distribution (142,254 normal training scores)
<code>models/lstm_ref_scores.npy</code>	Sorted LSTM reference distribution (99,454 normal training sequences)
<code>scripts/compute_reference_scores.py</code>	Computes and saves reference distributions from trained models
<code>scripts/calibrate_thresholds.py</code>	Runs fusion on validation set, prints score distributions and threshold analysis
<code>scripts/add_account_opening_date.py</code>	Adds <code>account_opening_date</code> to <code>clients.csv</code> with archetype-specific ranges
<code>tests/test_fusion.py</code>	End-to-end fusion test on test set
<code>tests/test_decision.py</code>	Full pipeline test: score $\rightarrow$ fuse $\rightarrow$ decide, with tier distribution and detection analysis

4 Next Steps

1. Fix batch pipeline: align PostgreSQL schema and profile computation with the rebuilt generator and 30-feature enrichment
2. SHAP integration in Scoring Service
3. Streaming pipeline via RedPanda (Enrichment $\rightarrow$ Scoring $\rightarrow$ Decision)
4. Streamlit dashboard consuming decisions topic